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Benchmarking deep learning architectures for hyperspectral in-vivo brain tumor segmentation

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ABSTRACT

Background and Objective: Deep Learning (DL) has become increasingly relevant in medical image analysis, especially for tasks such as surgical guidance. In this respect, Hyperspectral Imaging (HSI) has shown encouraging results for *in-vivo* brain tumor segmentation during surgeries, where DL models can benefit from spectral information. The literature includes a large collection of Neural Network (NN) architectures designed to process spatial and spectral information in HS images. However, most of these architectures have been developed outside the healthcare domain, and comparative studies on the main architectures applied to HS medical datasets are still limited.

Methods: In this work, we provide a systematic benchmark of the principal architectures that have emerged in the last five years, using the 25-band SLIMBRAIN and 128-band HELICoiD datasets. We have also established a taxonomy based on the mechanisms used to build the DL architectures, such as convolutions, attention, and a combination of both.

Results: Some of the convolutional models emerged as the best performing approaches, reaching a maximum F1-score of 65.08% and 91.13% on the two datasets, respectively, followed by hybrid and attention-based ones. Several compact architectures, such as DBDA and RSSAN demonstrated favorable accuracy-efficiency characteristics, showing that high performance is achievable across different spectral dimensionalities without large models.

Conclusions: Beyond providing a comparative ranking, this study identifies architectural mechanisms that are particularly relevant for medical HSI segmentation, including the role of Depth-Wise (DW) convolutions and spatial–spectral feature separation. These findings provide practical design indications for future HSI-based applications in neurosurgery and other medical domains.

1. Introduction

Gliomas are among the most common and aggressive forms of brain tumors, with a low survival rate and a significant impact on patients' health [1]. The annual incidence of brain tumors in the United States is 23.79 per 100,000 individuals, of which 29.80% are malignant in general, and 14.50% are Glioblastoma Multiforme (GBM), the most aggressive type [2]. These statistics are projected to increase as industrialization, urbanization and aging populations increase [3]. Surgical resection is the main treatment option. However, manual detection is prone to human error and the infiltrative characteristics of GBM [4] cause tumor tissue to extend beyond visible limits. Automatic detection systems, based on Artificial Intelligence (AI), could improve surgical results, reducing both the risk of leaving residual tumor cells and the removal of healthy tissue.

Medical imaging plays an important role in the detection and diagnosis of brain tumors [5]. However, automated image analysis such as that for MRIs (Magnetic Resonance Images) [6] can be complex due to variability in data, resolution, tumor morphology, and boundary ambiguity, which have motivated the development of increasingly advanced DL architectures for brain tumor segmentation [7]. In addition, these images cannot be easily acquired during surgery. A promising alternative is HSI [8,9]. It extends the conventional RGB technique by capturing reflectance information across multiple wavelength bands

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beyond the visible spectrum [10], thus allowing detailed insights into tissue composition. Moreover, HSI enables non-contact, non-ionizing image acquisition, making it particularly suitable for intraoperative use without interfering with the surgical workflow. This technology was originally developed for remote sensing applications where it was used, for example, to detect asbestos on rooftops [11]. Other fields where HSI has found increasing relevance are agriculture, environmental monitoring, and food quality assessment [12].

Recent studies have increasingly investigated the integration of HSI with DL models for the detection of gliomas [13], but few are assessing the latest architectures such as transformers [14,15], and, to the best of the authors' knowledge, none have considered more advanced models based on a hybrid convolution-attention mechanism, which have shown remarkable success in HS remote sensing tasks [16]. Furthermore, it appears important to analyze not only the performance of recent models, but also variables, such as spectral resolution and sensor type, that affect the system in real clinical settings.

In fact, HS cameras can capture up to hundreds of narrow spectral bands, but acquisition times may vary significantly depending on the setup. For instance, snapshot cameras can acquire an entire HS datacube in a single exposure of a few milliseconds, making them suitable for dynamic or time-constrained clinical scenarios, while push-broom scanners collect data line by line, typically offering higher spectral information at the cost of longer acquisition times. In both cases, depending on the number of spectral bands, processing such high-dimensional data during *in-vivo* brain tumor resection can pose a trade-off between computational speed and performance of tumor tissue detection. As a result, robust pre-processing techniques and domain-adapted DL algorithms are essential to translate HSI into practical tools [17]. Accordingly, this study compares the performance of several DL models employing two *in-vivo* datasets, namely the SLIMBRAIN database [18] and the HELICoiD dataset [19]. The former consists of more than one hundred subjects featuring 25 spectral bands, acquired using a snapshot camera. The latter contains fewer subjects with 128 bands acquired using a pushbroom linescan device.

Despite several recent systematic reviews summarizing DL approaches for brain tumor analysis [13,20], the domain of brain tumor segmentation with medical HSI still lacks comprehensive systematic DL benchmark studies. To the best of our knowledge, [21] conducted a benchmark study on HELICoiD, considering machine learning algorithms and a two-layer Convolutional Neural Network (CNN), achieving a median F1-score of $70.2 \pm 7.9\%$. Similar NNs were assessed in [22] for the initial version of the SLIMBRAIN database, achieving overall accuracy values between 60% and 95%. In contrast, other fields, such as remote sensing, have already seen the adoption of rigorous benchmarking methodologies based on the use of the PRISMA methodology (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [23]. Thus, having recognized these limitations, the contributions proposed in this work can be summarized as follows:

- a systematic cross-dataset evaluation of recent DL architectures for in-vivo brain tumor HSI segmentation, including models originally designed for remote sensing and RGB image analysis, under a common training and evaluation protocol, spanning markedly different spectral dimensionalities (namely, 25-band and 128-band data);
- a mechanism-oriented taxonomy of HSI NN, grouping models by convolutional, attention-based, and hybrid processing, and by spatial–spectral feature handling;
- an architectural analysis linking segmentation performance to specific design mechanisms, including depth-wise convolutional components, differentiated spatial–spectral processing, and the limitations of purely attention-based models;
- an accuracy-efficiency analysis considering F1-score, inference time, and model size, aimed at identifying architectures that are not only accurate but also suitable for practical intraoperative HSI applications.

Beyond these contributions, the techniques and models evaluated in this benchmark have significant potential for application in other intraoperative scenarios in which HSI can be used to define tumor limits [24].

The remainder of this paper is structured as follows. Section 2 describes the criteria chosen to select the models and the resulting taxonomy. In addition, it analyzes the datasets used and describes the training strategy. Section 3 presents the results which are later discussed in Section 4. Finally, Section 5 draws the conclusions.

2. Benchmark design and experimental protocol

A central objective of this benchmark was to compare heterogeneous DL architectures within a coherent and reproducible experimental framework. Representative models were therefore systematically identified, organized into an interpretable taxonomy, and evaluated on common *in-vivo* brain HSI datasets using the same training and testing conditions. This section describes the model selection process, the datasets used, and the unified experimental protocol.

2.1. Model selection and taxonomy

Given the limited number of studies on recent DL models for HSI-based brain tumor segmentation, candidate architectures were selected following the PRISMA methodology. The process began with a search and selection of relevant models, followed by an architectural analysis to understand their applicability to our work. Since this work benchmarks re-trained models on common medical HSI datasets rather than comparing the outcome reported in the original studies, PRISMA items 10–13 were not considered.

2.1.1. Eligibility criteria

Since many HSI architectures could be applied to brain tumor segmentation [25], restrictive eligibility criteria were required to make the benchmark feasible and reproducible.

We focused on original DL architectures designed for patch-based HSI classification or segmentation and built using convolutional layers (whether 2D or 3D), attention-based layers, or their combination. Accordingly, the literature search was conducted using the following Web of Science [26] queries:

```
TS=((hyperspectral image) AND (classification OR
detection) AND convolutional NOT graph NOT
transformer NOT attention NOT mamba) AND
PY=(2020-2025)

TS=((hyperspectral image) AND (classification OR
detection) AND (transformer OR attention) NOT
convolutional NOT graph NOT mamba) AND
PY=(2020-2025)

TS=((hyperspectral image) AND (classification OR
detection) AND (transformer OR attention) NOT
graph NOT mamba) AND PY=(2020-2025)
```

The first query targeted convolution-based models used in the HSI field for classification or detection tasks. This was achieved through the logical expression "`(hyperspectral image) AND (classification OR detection) AND convolutional`", while excluding keywords associated with other architectural families (graph, transformer, attention, and mamba). Similarly, the second query focused on pure attention-based models by requiring the presence of "`transformer OR attention`" and explicitly excluding "`convolutional`". The third query was instead constructed to capture hybrid convolution–attention architectures, which combine both processing mechanisms.

All searches were restricted to 2020–2025. Emerging approaches, such as Mamba models [27], were not included in the benchmark due to their still limited consolidation in the literature, as discussed among the limitations of this study.

Table 1
List of selected architectures grouped according to the proposed taxonomy: (a) convolutional, (b) attention, (c) hybrid and (d) *baseline* architectures.

	Model alias	Year	Original-study patch size	Original-study datasets	Source code implementation
(a)	Hyb3D2D [38]	2021	15 × 15	IP, PU, SV	Implemented following paper specs
	GhostNet [39]	2021	11 × 11	IP, KSC, PU, SV, UH	Implemented by third party researchers
	LiteDwNet [40]	2022	9 × 9	BS, IP, PC, PU, UH	Implemented following paper specs
(b)	SpectralFormer [41]	2022	9 × 9	IP, PU, UH	Study repository
(c)	DBDA [42]	2020	9 × 9	BS, IP, PU, SV	Study repository
	HiT [43]	2022	15 × 15	IP, PU, UH, XA	Study repository
	SSAN [44]	2020	7 × 7	IP, PU, SV	Study repository
	RSSAN [45]	2021	7 × 7	IP, KSC, PU	Implemented following paper specs
(d)	ViT [29]	2021	16 × 16	CIFAR, ImageNet, JFT	Implemented following paper specs
	EfficientNet [28]	2021	–	Cars, CIFAR, Flowers, ImageNet	Pytorch website implementation

2.1.2. Screening of the studies

According to the PRISMA flowchart in Fig. 1, publications were first filtered by year and citation count to retain recent and sufficiently established models. The citation-based threshold was adopted to prioritize architectures with sufficient methodological validation, visibility in the literature, and implementation details to support reproducible benchmarking. At the same time, this criterion may have limited the inclusion of very recent architectures, as further discussed in Section 4.4. Surveys, systematic reviews, works addressing tasks other than classification or segmentation, multimodal approaches (e.g. using additional information such as Light and Detection Ranging data), and methods not compatible with fixed-size patch-based inputs (e.g. superpixel-based methods) were excluded. Additional exclusions concerned architectures requiring multi-step training or transfer learning, and models originally designed for substantially larger patch size, which would have introduced heterogeneous spatial context across architectures. The procedure led to eight HSI architectures. To broaden the comparison, two widely used RGB image classification *baselines*, EfficientNetV2 [28] and Vision Transformer (ViT) [29], were also included because they satisfy the same layer-type constraints and are frequently used as backbones for image analysis tasks. All the final selected models are summarized in Table 1.

Notably, with the exception of the *baseline* methods, all selected architectures were originally designed for remote sensing datasets, including Botswana (BS), Indian Pines (IP), Kennedy Space Center (KSC), Pavia Center (PC), Pavia University (PU) and Salinas Valley (SV) [30], as well as University of Houston (UH) [31] and Xiongan (XA) [32]. This reflects the predominance of remote sensing in HSI research and explains why most architectures that meet the selection criteria originate in that domain. The *baseline* methods were designed for RGB datasets such as Cars [33], CIFAR [34], Flowers [35], ImageNet [36], and JFT [37], and were included because of their widespread use.

2.1.3. Taxonomy of the selected models

The selected architectures were grouped according to the dominant processing mechanism and the organization of spatial–spectral feature extraction, as summarized in Fig. 2.

Among HSI models, convolutional models included single-branch designs, such as the Hybrid 3D2D CNN [38] and GhostNet (Ghost Network) [39], and multi-branch designs, such as LiteDwNet (Lite Depthwise Network) [40]. Attention-based HSI models were represented by SpectralFormer [41]. Hybrid convolution-attention models included RSSAN (Residual Spectral-Spatial Attention Network) [45], HiT (Hyperspectral Image Transformer) [43], SSAN (Spectral-Spatial Attention Network) [44], and DBDA (Double-Branch Dual-Attention) model [42], and were further categorized according to whether the input layer was attention-based or convolutional, and whether spatial and spectral features were processed jointly or in separate stages.

Baseline RGB models were treated as a separate category, including EfficientNetV2 [28] as a convolutional architecture and ViT [29] as a transformer-based architecture.

This resulting taxonomy was used as the interpretative framework for the subsequent analyses, allowing the results to be discussed not only in terms of individual model rankings, but also according to broader architectural mechanisms.

2.2. In-vivo HSI datasets and exploratory analysis

In this benchmark, two publicly available HS datasets were used: the SLIMBRAIN database [18] and the HELICoiD dataset [19], differing significantly in terms of spatial and spectral resolution and acquisition setup.

2.2.1. SLIMBRAIN database

The SLIMBRAIN database [18] represents one of the most extensive and novel *in-vivo* HS databases of brain images, with more than 193 acquisitions. In this study, only 85 images showing clear scenes of the brain cortex were selected. The camera used was a first-generation MQ022HG-IM-SM5X5 Ximea snapshot camera, with a 35 mm Edmund Optics visible and near-infrared (VNIR) lens, capable of capturing 25 spectral bands in the range of 660–950 nm with a spatial resolution of 409 × 217 pixels. The camera was equipped with a high-pass optical filter with a cut-off wavelength of 650 nm to eliminate low-wavelength harmonics and reduce spectral noise. A 150 W halogen lamp connected to a fiber optic was used for illumination, ensuring uniform lighting in the surgical field. The exposure time ranged from 60 to 110 ms depending on the capture scene. In this work we followed the same preprocessing detailed in [18], including spectral calibration, using a white reference followed by a spectral correction to suppress the cross-talks that typically appear in Fabry-Pèrot-based filters.

2.2.2. HELICoiD dataset

The HELICoiD dataset was built in two subsequent acquisition campaigns [19,21] and represents the first publicly available collection of *in-vivo* HS images of human brain tissues. It was designed to support the development of AI-based tissue classification and segmentation methods. To date, it consists of 70 images from 40 subjects, acquired using a HS VNIR A-Series pushbroom scanner, equipped with a Xenoplan 1.4 lens, covering the 400–1000 nm spectral range with a spectral resolution of 2 to 3 nm, and producing up to 826 spectral bands per line. The maximum spatial resolution is 741 × 1004 pixels. The applied pre-processing pipeline was described in [21] and includes a spectral calibration using a white reference to obtain diffuse reflectance and the removal of the most noisy regions of the spectrum, centering the effective spectral range between 440.5 and 909.1 nm. Then, given the redundancy between adjacent bands, the 826 channels were reduced to 128 spectral bands. Similarly to the previous dataset, the illumination system was based on a quartz-tungsten halogen lamp connected to a cold light emitter through fiber optics to ensure homogeneous lighting in the 400–2200 nm spectral range. The scanning time varied depending on the number of lines scanned, with a maximum time of 60 s. Notably, the use of pushbroom technology, while enabling high spectral resolution, implies relatively long scanning times, highlighting the trade-off between spectral resolution and clinical usability.

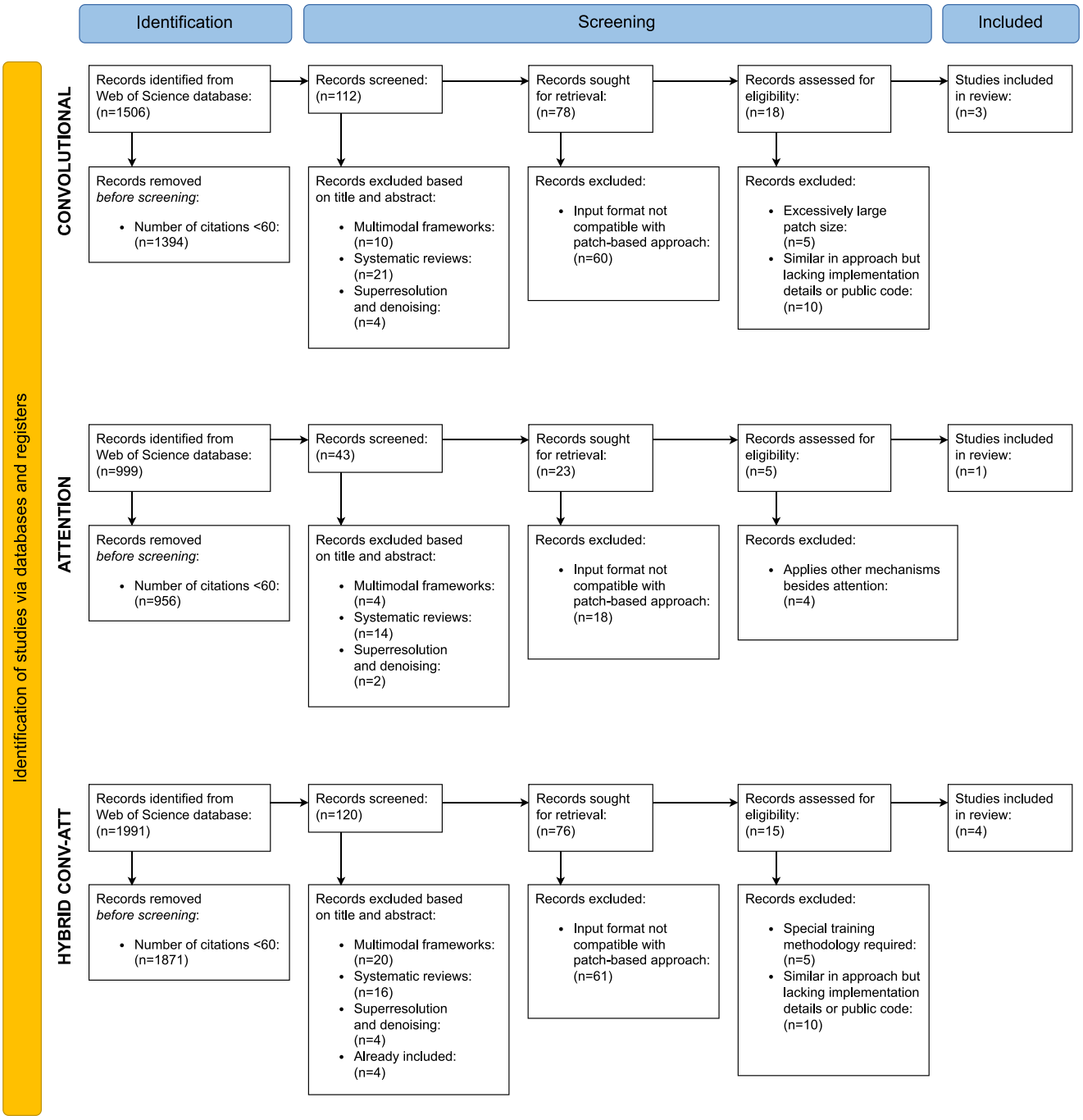


Fig. 1. PRISMA flowchart detailing the selection criteria for the three Web of Science queries.

2.2.3. Data comparison and exploration

For both datasets, Ground Truths (GTs) were obtained using a semi-automatic labeling process based on the spectral angle mapper [19]. As a result, four different classes were annotated: healthy tissue, tumor tissue, blood vessels, and dura mater (or background in the case of HELICoiD). As can be seen in Table 2, both datasets show an important class imbalance due to the natural distribution of tissues in a brain. The majority of annotated pixels in the two datasets belong to healthy tissue and dura mater (or background), followed by blood vessels and tumor tissue.

Regarding the type of tumor, both datasets feature primary gliomas ranging from grade I to grade IV, with the latter being the most prevalent in terms of number of images. In addition, both datasets include a limited number of secondary tumors resulting from metastases. Some

Table 2
Number of annotated pixels per class for both HS datasets.

Class	SLIMBRAIN	HELICoiD
Healthy	186 424	323 151
Tumor	32 166	42 288
Blood vessels	19 665	131 704
Dura mater/Background	90 284	523 574
Total	328 539	1 020 717
Images with labeled tumor	44	36
Images without labeled tumor	41	34
Total	85	70

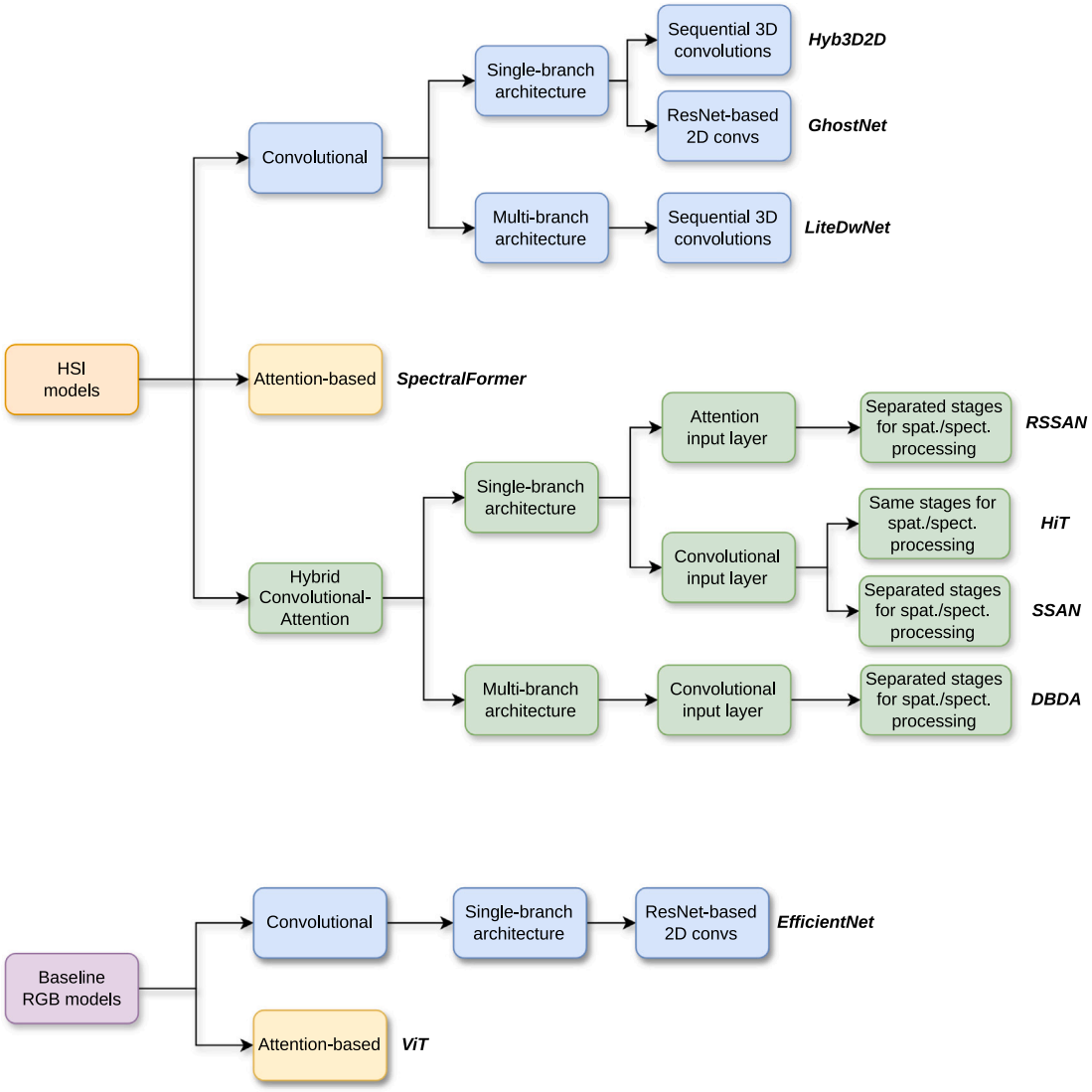


Fig. 2. Taxonomy of the selected models based on architectural family, branch structure, input layer, and spatial–spectral processing strategy.

images do not contain tumor/labeled pixels because the subjects underwent surgery for non-tumor conditions or because the tumor was not superficial and, therefore, was not captured in the image. In the case of SLIMBRAIN, 53 subjects with gliomas of various grades, 4 had multiple grade astrocytomas, 6 had meningioma and 1 had oligodendroglioma. Six subjects had secondary tumors, while the remaining 15 underwent surgery for cerebrovascular diseases. In HELICoiD, 30 out of 40 subjects were diagnosed with primary tumors, including 5 grade I, 6 grade II, 3 grade III, and 16 grade IV cases. The remaining subjects included 7 cases with secondary tumors and 3 cases without brain tumors.

To visually understand how distinguishable the spectral signatures of different tissue classes are before model training, we performed an analysis of class separability using the t–distributed Stochastic Neighbor Embedding (t-SNE) algorithm [46]. This technique takes high-dimensional data, such as the 25 or 128 spectral bands of the two HS datasets in this study and projects them into a two-dimensional space while maintaining local proximity. To accelerate this computation, a subset of 10% of the datasets was selected while maintaining the same class proportion. As a result, Fig. 3 shows that in SLIMBRAIN, healthy tissue and dura mater are the main clusters formed, even if they partially overlap. Tumor points are in between the healthy tissue and dura mater clusters, mostly overlapped. This may suggest a potential challenge for DL algorithms in clearly distinguishing between the four classes. Similarly, blood vessels are represented very close to the dura mater or healthy tissue points. In contrast, Fig. 3 also shows that in HELICoiD three more distinct clusters are generated for healthy tissue, background, and blood vessels. However, also in this case, some tumor points are partially overlapped with other clusters, albeit with some small isolated groups that are potentially easier to classify.

To further examine the cluster differences observed in the t-SNE embeddings of SLIMBRAIN and HELICoiD, Fig. 4 reports the average signal-to-noise ratio (SNR) for each spectral band. To obtain a more representative estimate, only the craniotomy site defined by the top-left and bottom-right coordinates of the GT samples in each image is considered, thereby excluding areas that do not contribute to the training process. The SNR was computed as $\mathrm{SNR(dB)} = 10\log_{10}\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right)$, where the signal power, P_{signal}, is calculated as the square of the mean value of the samples within the selected region for a given band. The noise power, P_{noise}, is obtained using the method proposed in [47] for noise-level estimation in textured images. The resulting SNR profiles provide additional support for the differences observed in the t-SNE projections, as SLIMBRAIN contains fewer spectral bands and exhibits substantially lower SNR values than HELICoiD. Together, these two

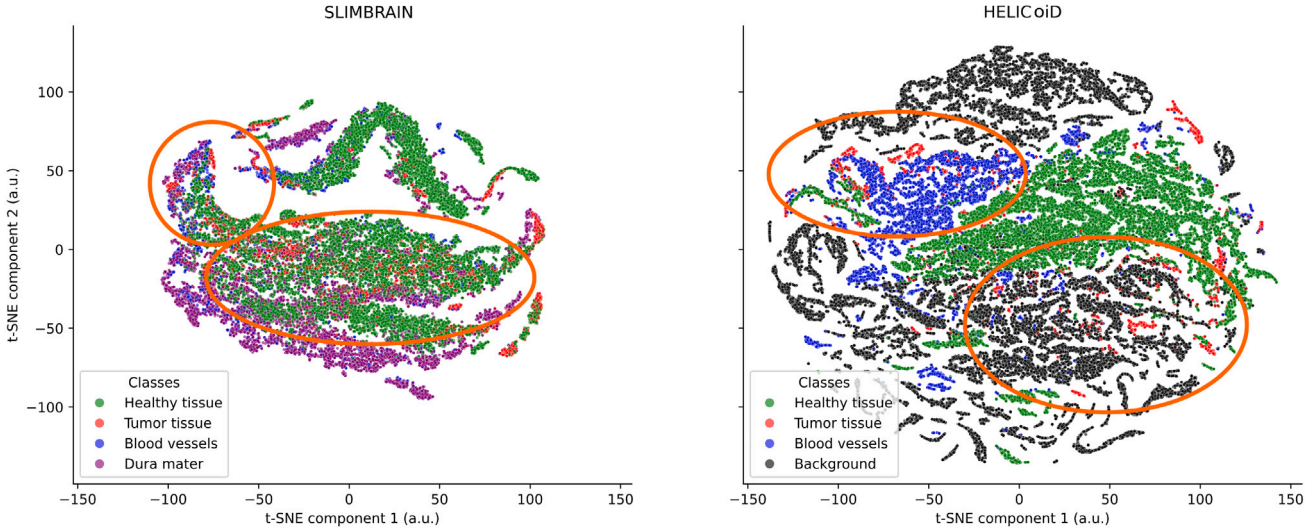


Fig. 3. Scatterplots of the class separability for the two datasets. Obtained using the t-SNE algorithm and a 10% stratified subsampling of the datasets. For both datasets, orange ellipses highlight the main overlapping region between classes.

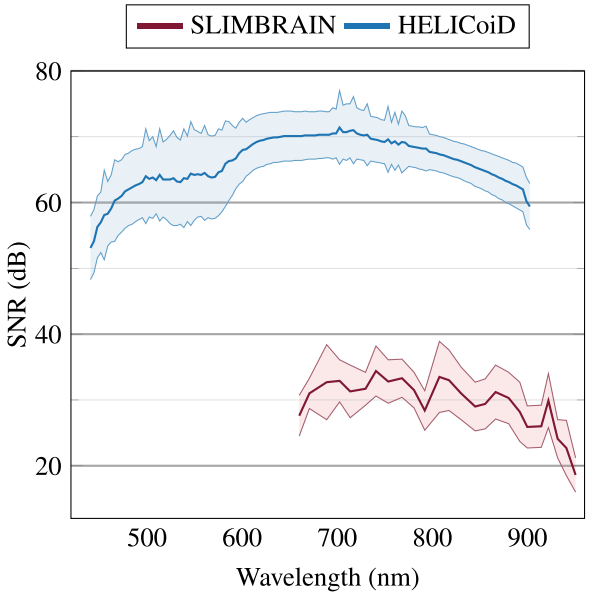


Fig. 4. Average SNR per spectral band for the SLIMBRAIN and HELICoiD datasets. Solid lines represent the mean SNR, while the shaded regions denote one standard deviation above and below the mean. The overall SNR is 29.62 ± 3.71 dB for SLIMBRAIN and 66.11 ± 3.83 dB for HELICoiD.

factors may contribute to the reduced separation between tissue classes observed in the projected feature space.

2.3. Experimental protocol

Following the definition of the selected architectures and datasets, a unified experimental protocol was adopted to ensure a consistent comparison across models. All architectures were trained and evaluated using the same data preparation procedure, patient-wise splitting strategy, patch size, training settings, and metrics. The following subsections describe data preparation, model training, and final evaluation.

2.3.1. Data preparation

Data preparation was performed independently for the two datasets, as summarized in Fig. 5. The workflow included tumor-aware patient-wise splitting, training-only GT augmentation, spectral normalization, and patch extraction.

To assess the generalizability of the models, we adopted a patient-wise data split, independently applied to SLIMBRAIN and HELICoiD without merging the two datasets. For each training-evaluation iteration, the dataset was first divided into images with and without tumor annotations. Each subset was then independently split into training (70%), validation (20%), and test (10%) partitions. The corresponding subsets were subsequently merged, as shown in Fig. 5, to ensure that training, validation, and test sets each contain at least one image with tumor annotations.

Class imbalance was addressed through two complementary strategies: GT augmentation and weighted loss function. First, morphological augmentation was applied only to the GT of the training set, while validation and test GTs were left unchanged. This operation did not modify the HS images themselves, but only increased the number of labeled training pixels belonging to under-represented classes. Specifically, for each training GT, the binary masks corresponding to tumor or blood vessels pixels were processed using a morphological closing operation, defined as a dilation followed by an erosion, with an elliptical kernel of size $k = 3$. The dimension of the kernel was selected to locally fit small gaps while preserving the overall morphology of the labeled regions. However, this procedure was intended to mitigate class imbalance at the annotation level and did not enforce a perfectly balanced class distribution. Therefore, as a second class-balancing strategy, we evaluated both the weighted cross-entropy loss and focal loss [48] functions, selecting the latter, with inverse-frequency weighting, for its better ability to emphasize under-represented classes.

To mitigate intrinsic variability in amplitude across spectral channels and stabilize the training process, in each pre-processed HS image, the spectral bands were independently normalized using Eq. (1), with the minimum and maximum band-wise values computed on the training and validation sets. In Eq. (1), $\tilde{x}$ is the normalized spectral signature, and x, $\tilde{x}$, x_{min}, and x_{max} are vectors in $\mathbb{R}^d$, with d denoting the number of spectral bands of the dataset.

$$\tilde{x} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{1}$$

Finally, for each training, validation and test set, a patching procedure was used to extract $p \in \mathbb{R}^{n \times n \times d}$ patches, centered on their

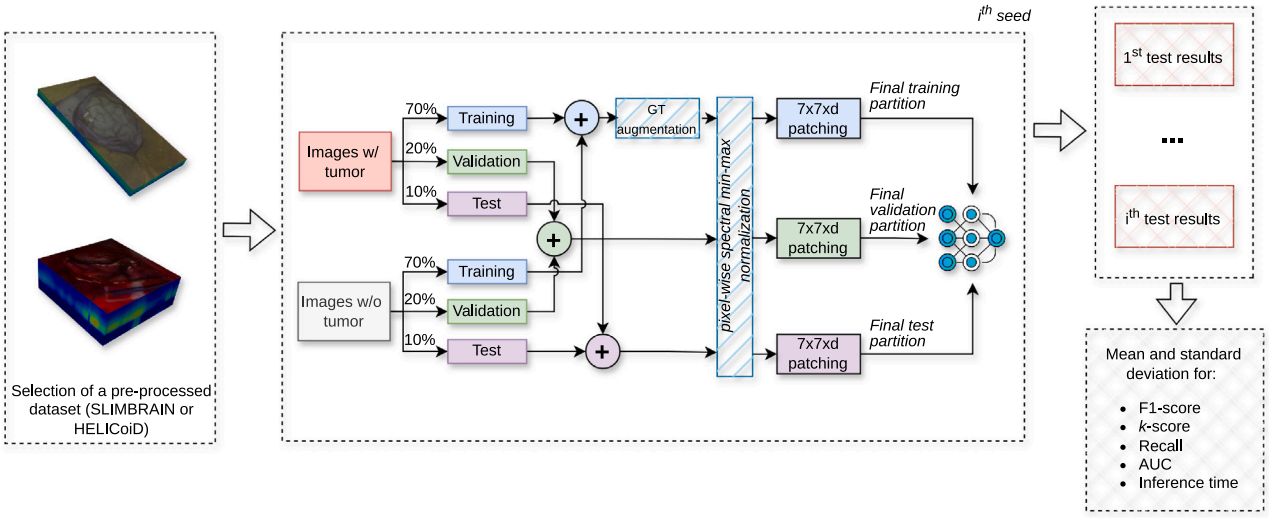


Fig. 5. Training and evaluation workflow with tumor-aware splitting, training-only GT augmentation, spectral normalization, patch extraction, and repeated-seed metric aggregation.

corresponding HS labeled pixel, where n is the spatial dimension and d the spectral one (e.g. 25 or 128). This yields a set of overlapping patches which were then used for model training and evaluation. To preserve the original height and width of the HS cube, zero padding of size $\lfloor n/2 \rfloor \times \lfloor n/2 \rfloor$ was applied. Previous studies on HS brain tumor imaging [49,50] often employed a large spatial context (e.g. 11×11 spectral pixels). However, preliminary tests revealed that such configurations significantly increased memory usage and computational time, particularly for the HELICoiD dataset, preventing the experiments from being completed within a reasonable timeframe. Thus, to balance spatial context and resource efficiency, a fixed patch size of 7×7 was adopted for all experiments.

2.3.2. Training strategy

All models were trained using a common optimization strategy to ensure comparable experimental conditions across architectures and datasets. The batch size was set to a relatively large 2 048 patches to accelerate model convergence and maximize GPU memory utilization. To counteract the potential performance degradation associated with large batches, the Layer-wise Adaptive Rate Control (LARC) optimizer [51] was used to adaptively scale the learning rate according to the model learnable parameter norms. In any case, the initial learning rate of 10^{-3} was dynamically adjusted according to a cosine annealing schedule that promotes smoother convergence while reducing the risk of suboptimal local minima. All models were trained for 300 epochs with the validation set evaluated every 10 epochs against the weighted global F1-score. At the end of the 300 epochs, the version of the model that obtained the highest F1-score was selected to perform the inference over the test set. For a fair comparison of the models, no optimization of the hyperparameters was performed. Moreover, to ensure statistical robustness, each model was evaluated five times using five independently randomized train, validation, and test splits. Randomization was performed using Python's pseudorandom generator (MT19937) with fixed seeds, one per iteration, to ensure the reproducibility of each partition.

The models were trained using the PyTorch framework [52], employing the distributed data parallel module to distribute each training load over four GPUs. The overall training hardware infrastructure consisted of 16 NVIDIA A16 vGPUs distributed across two nodes, each equipped with two Intel Xeon Silver CPUs and 768 GB of DDR4 RAM. Training progress was tracked using the MLflow framework [53]. On the other hand, for the final inferences, a system equipped with an NVIDIA GeForce RTX 4090 GPU, an Intel Core i9-14900KF CPU, and 128 GB of RAM was used to emulate the computational capabilities of a potential clinical device.

2.3.3. Evaluation and comparison criteria

Model evaluation was designed to assess both segmentation accuracy and computational efficiency, reflecting the dual requirement of reliable tissue identification and practical deployability in intraoperative HSI scenarios. As summarized in Fig. 5, all metrics were computed on the test set for each fold and reported as mean and standard deviation.

In terms of classification performance, given the clinical importance of correctly identifying each tissue, the quantitative analysis focused on metrics such as Cohen's κ-score, weighted recall, F1-score, and area under the receiver operating characteristic curve (AUC) as one-vs-the-rest. When focusing on individual classes, these metrics were computed per class. In particular, in this case, the AUC was weighted by class frequency.

To assess whether the differences between competing models were statistically significant, pairwise Wilcoxon signed-rank tests were performed following the methodology proposed in [54]. The test was applied to the per-image mean F1-score, obtained by averaging the F1-score across the five folds for each test image. For each dataset, the models were first ranked according to their average F1-score. Then, for each pair of adjacent models in the ranking, the null hypothesis stated that the higher-ranked model did not achieve a statistically significant improvement over the immediately lower-ranked model across the test examples. Additionally, for the top three performing models, the Wilcoxon test was also conducted against the models located two positions below them in the ranking.

Regarding computational efficiency, the total number of trainable parameters P of each architecture was considered as an indicator of their complexity and memory footprint. Additionally, to evaluate the scalability and processing time requirements, the average inference time per patch t was calculated as the time required to predict the complete test set divided by the total number of predicted patches. Since all pixels in the test images were used to extract patches and the HELICoiD dataset includes images of varying height and width, this measure provides an estimate of the expected inference time per HS cube for arbitrary image sizes. Notably, these performance metrics depend on the hardware used.

Finally, to support the interpretation of architectural behavior, feature separability was assessed using the Supervised Discriminant Isomap (SDI) algorithm [55]. SDI was used as an auxiliary analysis tool, rather than as a primary performance metric, to examine tissue-class separation in the learned representations. For each model, SDI was computed at selected intermediate outputs during forward passes

on the five-fold test populations, and the values averaged. Input-data SDI was used as a reference, whereas final-logit SDI assessed the separability induced by the complete model.

3. Results

The results are organized to progressively support the interpretation of the benchmark. Global and class-specific metrics are first reported to compare the segmentation performance of the selected architectures across the two datasets. Computational indicators, including model size and inference time, are then analyzed together with F1-score to evaluate accuracy-efficiency trade-offs. Qualitative examples are finally presented to complement the numerical evaluation and illustrate representative segmentation outcomes.

3.1. Segmentation performance and model ranking

Fig. 6 shows the model ranking on the two datasets according to the average F1-score across the five folds. The corresponding statistical significance analysis, performed as described in Section 2.3.3, is also reported in the figure. The *p*-values indicate the pairwise Wilcoxon signed-rank tests between adjacent models in the ranking, with descending arrows identifying the compared pairs. For the top three performing models, additional comparisons with the models located two positions below them are also shown.

SLIMBRAIN generally performs worse than HELICoiD, most likely due to the class overlap mentioned in Section 2.2.3, causing the models to have difficulty discriminating between tissues. Surprisingly, the best performing model with SLIMBRAIN is EfficientNet, despite being originally designed for RGB images. However, the huge number of parameters for this network and the limited 65.08% F1-score easily justify that a model designed for RGB acquisitions can achieve a performance not far from models specifically designed for HS images. Nonetheless, the Wilcoxon signed-rank test performed between EfficientNet and Hyb3D2D yields a *p*-value of 0.104, indicating that the improvement obtained by EfficientNet is not consistently observed across the SLIMBRAIN test examples.

Hyb3D2D and DBDA achieved second and third place on SLIMBRAIN and first and second place on HELICoiD, respectively, showing a very similar performance in both datasets. The slight difference in overall F1-score in favor of Hyb3D2D corresponds to marginal improvements, as reflected by the *p*-values of 0.428 and 0.284 obtained for the SLIMBRAIN and HELICoiD datasets, respectively. Consequently, no statistically significant performance difference can be established between Hyb3D2D and DBDA on either dataset. However, both models demonstrate statistically superior performance compared with the third-ranked model on the HELICoiD dataset, RSSAN. Despite their different architectures, Hyb3D2D and DBDA showed consistent rankings across the two datasets, suggesting robust behavior within the experimental setting considered in this benchmark. The possible architectural factors underlying this behavior are discussed in Section 4.1. Moreover, consistency across datasets suggests that they are among the most robust HSI models in this benchmark.

On HELICoiD, EfficientNet and RSSAN achieve a notable F1-score, providing evidence that they can be considered suitable for spectral data, including medical applications. However, the *p*-value of 0.829 obtained between EfficientNet and SSAN indicates that, despite its lower mean F1-score, SSAN does not perform consistently worse than EfficientNet across the test dataset, further supporting SSAN as a competitive alternative for HS tissue segmentation. Another notable observation is the relative improvement of RSSAN from SLIMBRAIN to HELICoiD, where it moved from a lower ranking position to the third-best result. In most cases, with the exception of GhostNet and LiteDwNet, architectures containing convolutional layers ranked above purely attention-based models. Among attention-based approaches, SpectralFormer consistently outperformed ViT on both datasets. These architectural trends are further discussed in Section 4.1.

Overall, the pairwise Wilcoxon tests conducted on the SLIMBRAIN results indicate that adjacent models in the ranking achieve relatively similar performance, with limited improvements between consecutive positions. Nevertheless, the results obtained by EfficientNet and Hyb3D2D position these architectures among the most promising approaches for hyperspectral tissue segmentation in this benchmark.

The rankings in Fig. 6 are consistent with the metrics reported in Tables 3 and 4. Since the tumor class is the most clinically relevant, both tables also report the average tumor F1-score across the five folds. This metric further confirms that models struggle to discriminate tumor samples in SLIMBRAIN, where tumor segmentation performance is markedly lower than in HELICoiD, although the latter also achieves only moderate tumor F1-scores. For SLIMBRAIN, the top ranked models in Fig. 6, EfficientNet and Hyb3D2D, also obtain the highest global metrics in Table 3. However, this trend is not fully preserved for the tumor class, reflecting the difficulty of segmenting this tissue. In general, the results show high variability, as reflected by their standard deviations. This variability can be due to the heterogeneity of the test images which include acquisitions related to different setup versions and distinct scenarios, leading to strong performance on some images and poor segmentation on others.

3.2. Accuracy-efficiency trade-off

After comparing segmentation performance, we analyzed whether the ranked models also provide favorable computational characteristics. Fig. 7a shows the trade-off between the number of parameters of each model, on a logarithmic scale, and the average F1-score. Two distinct clusters emerge due to the difference in F1-score obtained with SLIMBRAIN and HELICoiD. Overall, most models show a slight variation in the number of parameters which depends on the number of spectral bands in use. In contrast, ViT and EfficientNet seem to maintain a constant number of parameters, because the input linear transformation makes the remaining architecture largely independent of the channel dimension. Generally, an increase in the number of parameters tends to result in a corresponding slow increase in the average F1-score. However, models with few parameters such as DBDA, RSSAN, and also SpectralFormer achieve a relatively satisfactory F1-score on both datasets. Hyb3D2D and EfficientNet resulted in the highest number of parameters, and also top performance. For the HELICoiD dataset, HiT achieves high F1-score values but, compared to Hyb3D2D, the many additional parameters reduce its attractiveness from an efficiency perspective. In contrast, LiteDwNet is the most efficient model on HELICoiD (85% F1-score and ~35K parameters) but fails to generalize effectively to the SLIMBRAIN setup. Notably, despite the computational burden associated with attention matrices, ViT and, by extension, SpectralFormer occupy a central position in the scatterplot.

Regarding inference time, Fig. 7b associates the average time required to classify a single 7×7 patch, on a logarithmic scale, with the average F1-score. Architectures trained on SLIMBRAIN appear to be faster compared to those trained on HELICoiD because of the lower dimensionality of the data, which is reflected in a reduced number of mathematical operations required by the algorithms. Computing the centroids of the two formed clusters, from the values reported in the tables, shows that processing SLIMBRAIN patches is approximately 2× faster than processing HELICoiD ones. Excluding HiT and EfficientNet, Fig. 7b shows that with SLIMBRAIN the models are clustered close together with little variability in inference time, making it difficult to distinguish an actual best performer. However, DBDA stands out with an F1-score of 61.83%, a fast inference time of 10.17 μs, and a compact model size (~60K parameters). On the other hand, in the case of HELICoiD, Hyb3D2D resulted in top classification performance with

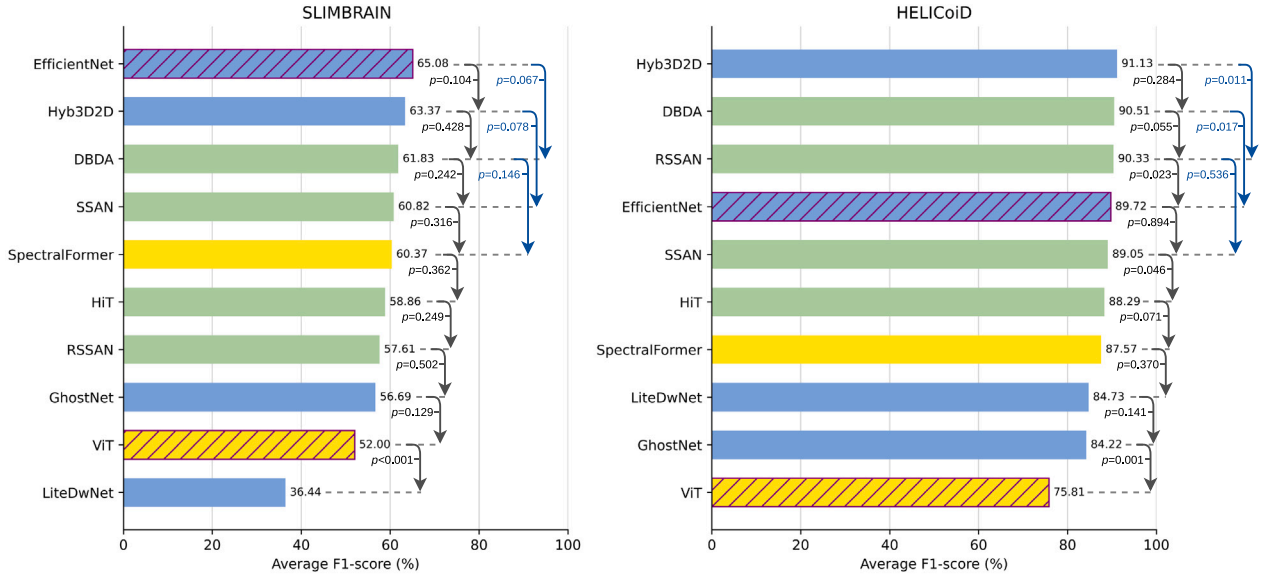


Fig. 6. Model ranking for the SLIMBRAIN and HELICoiD datasets according to the average F1-score across the five folds. The color of each bar matches the corresponding taxonomy in Fig. 2. Hatched bars indicate models originally designed for RGB image segmentation. Pairwise *p*-values are reported on the right side of each bar, with arrows indicating the compared model pairs and blue arrows highlighting comparisons with the best-performing model.

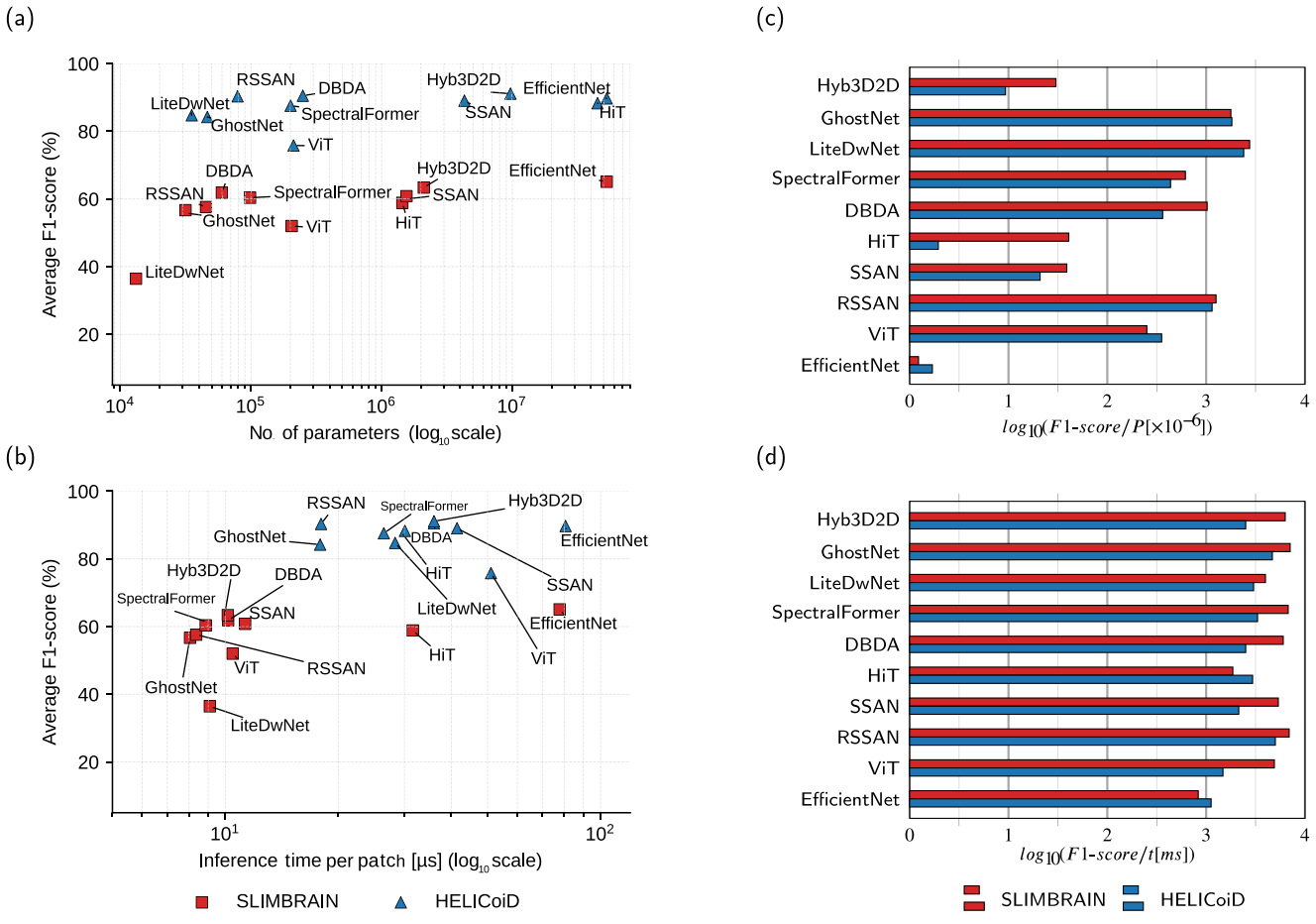


Fig. 7. Analysis of (a) Average F1-score as a function of the number of trainable parameters P, shown on a logarithmic scale. (b) Average F1-score as a function of the average inference time per patch t, also shown on a logarithmic scale. (c) Parameter efficiency, computed as the logarithmic ratio between the average F1-score and the number of trainable parameters. (d) Computational efficiency, computed as the logarithmic ratio between the average F1-score and the average inference time per patch.

Table 3
Average results (± standard deviation) of all models for the considered metrics, computed on the SLIMBRAIN database.

Model	SLIMBRAIN						
	F1-score (%)	κ-score (%)	Recall (%)	AUC (%)	F1-score tumor class (%)	Inference time per patch (μs)	No. of parameters
Hyb3D2D	63.37 ± 21.98	40.15 ± 22.80	**64.47 ± 20.26**	**84.92 ± 12.21**	12.14 ± 25.08	10.15 ± 2.93	2 102 628
GhostNet	56.69 ± 22.25	30.87 ± 21.77	57.10 ± 20.81	77.19 ± 16.08	16.24 ± 22.38	**8.05 ± 1.03**	31 554
LiteDwNet	36.44 ± 25.36	19.56 ± 18.51	36.31 ± 22.96	83.23 ± 15.10	17.57 ± 20.18	9.09 ± 4.66	13 180
SpectralFormer	60.37 ± 22.50	35.91 ± 23.99	60.05 ± 21.36	80.98 ± 16.11	13.11 ± 19.14	8.86 ± 0.76	98 450
DBDA	61.83 ± 22.61	38.42 ± 25.75	62.84 ± 21.20	82.36 ± 14.40	11.08 ± 23.13	10.17 ± 2.23	59 895
HiT	58.86 ± 23.33	35.38 ± 23.79	57.97 ± 22.14	82.41 ± 15.88	22.40 ± 23.52	31.58 ± 3.17	1 441 569
SSAN	60.82 ± 24.02	37.80 ± 25.08	61.38 ± 22.00	81.23 ± 16.98	11.68 ± 23.02	11.29 ± 2.99	1 550 276
RSSAN	57.61 ± 22.77	33.23 ± 21.89	57.46 ± 21.45	78.86 ± 15.73	17.02 ± 26.14	8.36 ± 0.87	45 287
ViT	52.00 ± 23.23	26.41 ± 22.18	51.20 ± 22.67	77.58 ± 14.65	**28.08 ± 21.53**	10.46 ± 0.99	205 252
EfficientNet	**65.08 ± 20.90**	**42.64 ± 23.44**	64.35 ± 20.17	83.02 ± 11.46	20.66 ± 26.53	77.79 ± 11.39	52 868 232

Table 4
Average results (± standard deviation) of all models for the considered metrics, computed on the HELICoiD dataset.

Model	HELICoiD						
	F1-score (%)	κ-score (%)	Recall (%)	AUC (%)	F1-score tumor class (%)	Inference time per patch (μs)	No. of parameters
Hyb3D2D	**91.13 ± 16.06**	**83.69 ± 16.68**	**90.22 ± 16.50**	**98.33 ± 3.77**	58.00 ± 36.29	35.98 ± 5.11	9 696 612
GhostNet	84.22 ± 18.79	71.58 ± 21.67	82.61 ± 19.72	95.51 ± 7.74	40.84 ± 33.80	**17.89 ± 1.14**	46 386
LiteDwNet	84.73 ± 18.82	70.29 ± 25.35	82.14 ± 20.71	97.85 ± 3.66	57.94 ± 34.96	28.34 ± 2.85	35 212
SpectralFormer	87.57 ± 12.98	72.72 ± 21.04	84.90 ± 15.15	97.67 ± 3.05	50.67 ± 33.86	26.44 ± 1.38	201 141
DBDA	90.51 ± 12.66	80.25 ± 20.18	88.61 ± 14.69	97.94 ± 5.25	60.53 ± 33.67	35.95 ± 5.19	249 567
HiT	88.29 ± 15.95	77.00 ± 20.83	86.51 ± 17.34	98.20 ± 2.99	55.50 ± 34.89	30.09 ± 2.34	44 960 609
SSAN	89.05 ± 17.99	81.25 ± 17.93	87.90 ± 18.65	98.02 ± 4.09	54.78 ± 36.75	41.43 ± 7.05	4 302 436
RSSAN	90.33 ± 14.07	79.95 ± 17.84	88.74 ± 15.49	98.16 ± 2.42	58.07 ± 34.71	17.99 ± 1.15	79 013
ViT	75.81 ± 22.14	56.77 ± 26.86	71.09 ± 23.63	94.83 ± 9.44	40.10 ± 26.29	51.03 ± 0.89	211 844
EfficientNet	89.72 ± 13.33	78.43 ± 18.33	87.80 ± 14.76	97.11 ± 3.48	**61.77 ± 32.78**	80.66 ± 2.28	52 890 480

medium inference time, while RSSAN and GhostNet are the fastest, with RSSAN achieving a performance close to that of Hyb3D2D.

Figs. 7c and 7d further summarize the efficiency of the models by relating the F1-score to the number of parameters and the inference time. Fig. 7c reports the value of $\log_{10}(F1\text{-}score/P[\times 10^{-6}])$, which expresses how much segmentation performance is obtained per million parameters, with higher values indicating a more favorable accuracy-per-parameter efficiency. This allows a direct comparison across architectures with different sizes. Confirming what was seen in Fig. 7a, LiteDwNet, GhostNet and RSSAN show the highest values on SLIMBRAIN and HELICoiD, demonstrating that their compact design provides a favorable F1-score–complexity balance. EfficientNet yields the lowest value due to its large number of parameters that do not justify its choice with respect to the other more efficient models. As already noted, since the number of parameters does not change drastically from SLIMBRAIN to HELICoiD, the trend is the same with the latter, except for the HiT, for which there is a steady increase in the number of parameters passing from SLIMBRAIN to HELICoiD, mainly because of the larger number of spectral bands, which results in 43M additional parameters. This increase comes from the higher number of weights and biases required in each permutator block to process the expanded feature set.

On the other hand, Fig. 7d shows the corresponding ratio $\log_{10}(F1\text{-}score/t[\text{ms}])$, where t is the inference time in milliseconds for a single 7×7 patch. The plot indicates that most architectures achieve comparable efficiency on both datasets, with EfficientNet exhibiting the lowest accuracy–time efficiency, while GhostNet and DBDA stand out as the most favorable models thanks to their relatively high F1-scores combined with very short inference times. Although the different spectral dimensionality between SLIMBRAIN and HELICoiD naturally affects the absolute values, the trends remained consistent with those observed in Fig. 7b.

3.3. Qualitative assessment

Since quantitative metrics can only be computed on labeled pixels and GT annotations are sparse, qualitative examples provide a necessary complement to the numerical evaluation.

The selected examples illustrate poor, average, and good segmentation cases according to the per-image F1-score quantiles computed over the complete set of test images across the five folds. For each test image in each fold, the mean F1-score is first averaged across the ten models considered in this study, resulting in a single representative score per image. The images are then ranked according to this score, and representative examples are selected from the 5th, 50th, and 95th percentiles of the distribution. As a result, Fig. 8 for SLIMBRAIN and Fig. 9 for HELICoiD indicate in the first column examples whose average segmentation performance across models is better than 5% of the test images ($F1_{05}$ row), 50% of the test images ($F1_{50}$ row) and 95% of the test images ($F1_{95}$ row), respectively.

3.3.1. SLIMBRAIN database

Images in Fig. 8 further confirm the difficulty of segmenting SLIMBRAIN images already highlighted in the quantitative analysis. The feature overlap between classes shown in Fig. 3 appears to have a direct impact on the resulting segmentation maps, particularly for the $F1_{05}$ and $F1_{50}$ rows. In these examples, the similarities between healthy and tumor tissue cause the three evaluated models to struggle to accurately

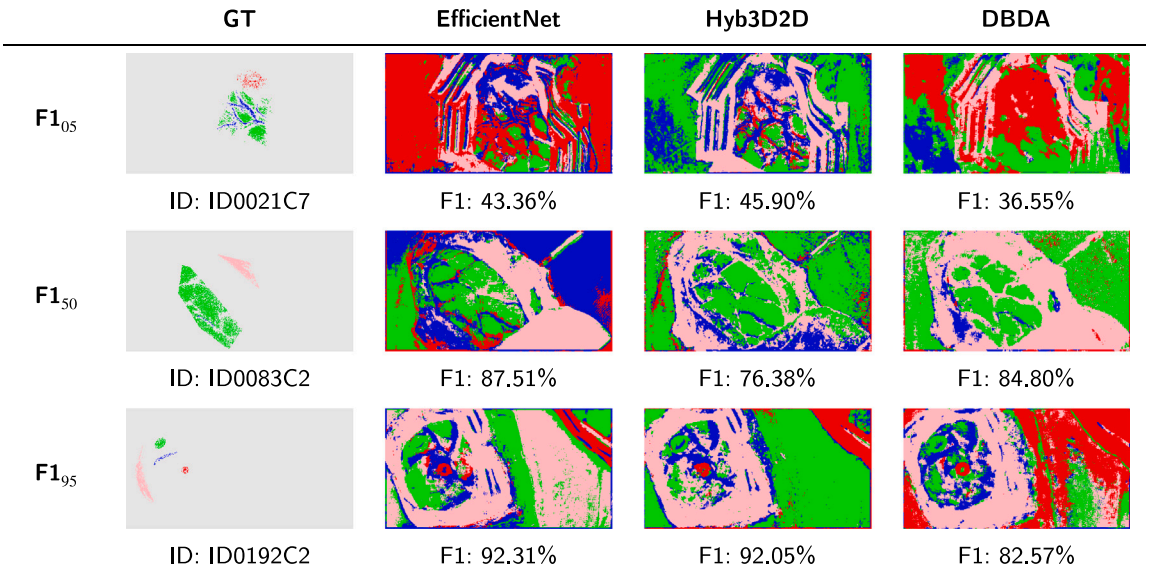


Fig. 8. Representative SLIMBRAIN examples for the three best-performing models presented in Fig. 6. Rows $F1_{05}$, $F1_{50}$, and $F1_{95}$ denote examples selected from the 5th, 50th, and 95th percentiles of the per-image F1-score distribution. Colors indicate healthy tissue in green, tumor in red, blood vessels in blue, and dura mater in pink.

identify tumor regions without overextending the segmentation into areas labeled as healthy tissue. The results corresponding to the $F1_{50}$ row further suggest that the models frequently confuse vascular structures with dura mater tissue. In contrast, the $F1_{95}$ row illustrates test cases in which the three models correctly identify most of the labeled tissue, despite the limited number of annotated pixels available for evaluation. The favorable metrics obtained for the $F1_{95}$ examples can be contrasted with both the large standard deviation reported in Table 3 and the poorer results observed for the $F1_{05}$ examples. This observation suggests that the lower overall F1-score obtained on the SLIMBRAIN dataset, compared with HELICoiD, may be largely attributable to the poor performance on hard test examples.

3.3.2. HELICoiD dataset

The segmentation maps obtained for the three best-performing models on the HELICoiD dataset are presented in Fig. 9. A first notable observation is that, in the $F1_{05}$ row, unlike in the SLIMBRAIN dataset, the Hyb3D2D and DBDA models are capable of correctly identifying healthy tissue and dura mater, which are the two most abundant classes in the dataset. This baseline segmentation capability, even in the most challenging examples, increases the minimum expected overall F1-score on the HELICoiD dataset. However, the $F1_{05}$ row also reveals a behavior previously observed in the SLIMBRAIN dataset: the misclassification of tumor tissue as dura mater. The segmentation maps corresponding to the $F1_{50}$ and $F1_{95}$ rows exhibit similar visual appearance and comparable F1-score values, suggesting that the three models are capable of effectively segmenting the four analyzed tissue classes for the majority of the test examples. Nonetheless, in all displayed cases, the models only achieve a coarse segmentation of the vascular tissue, with limited detail in the thinner vessels. This limitation may be attributable to the poor definition of vessel contours in the images and annotations.

4. Discussion

The benchmark results show that the suitability of DL architectures for brain HSI segmentation cannot be fully explained by quantitative rankings alone. Instead, model behavior should be interpreted by considering how each architecture extracts and combines spectral-spatial information, how stable its predictions remain across images and datasets, and how computational demands and prediction variability may affect reliability in clinical scenarios. All aspects discussed in the following.

4.1. Architectural behavior and spectral-spatial feature extraction

When interpreted through the taxonomy introduced in Section 2.1.3, the results suggest that the most consistent models were those combining strong local inductive biases with an explicit or implicit separation of spatial and spectral feature processing. Although no architectural family was uniformly superior across both datasets, convolutional and hybrid models generally performed more consistently than purely attention-based ones.

The performance differences were further investigated through the architectural behavior shown in Fig. 10. Each evaluated model is represented in terms of the fundamental building blocks implemented in their constituent layers. The upper part of Fig. 10 marks with red dots the points at which SDI is computed, while the lower part reports the corresponding values for each dataset and model. As described in Section 2.3.3, higher SDI values indicate better class separability in the learned feature space.

Pure attention-based architectures, such as ViT and SpectralFormer, achieved lower performance than the best convolutional and hybrid models. This suggests that the attention mechanism alone may be less effective in capturing local spatial patterns than convolutional operations. However, the performance gap between SpectralFormer and its *baseline* counterpart, ViT, indicates that architectural adaptations specifically designed for HSI data, such as GSE (Group-wise Spectral Embedding) and CAF (Cross-layer Adaptive Fusion), can partially mitigate this limitation. In particular, Fig. 10 shows that SpectralFormer exhibits an increase in SDI metric for both HELICoiD and SLIMBRAIN immediately after applying the CAF module, suggesting that this component improves the separability of the learned representations.

A similar but more nuanced behavior can be observed for RSSAN, whose attention-based header processes spectral and spatial information separately through its spectral and spatial attention stages. RSSAN also shows that attention-based input processing can be effective, although its different ranking across datasets suggests stronger dependence on spectral richness. In particular, the lower spectral sampling of SLIMBRAIN may have limited the effectiveness of its attention-based input header compared to HELICoiD. This interpretation is further supported by the SDI progression across measurement points. For the HELICoiD dataset, RSSAN shows a rapid increase in SDI just before the final Feed-Forward Network (FFN) stage, after maintaining relatively low values at the previous measurement points. This behavior can be

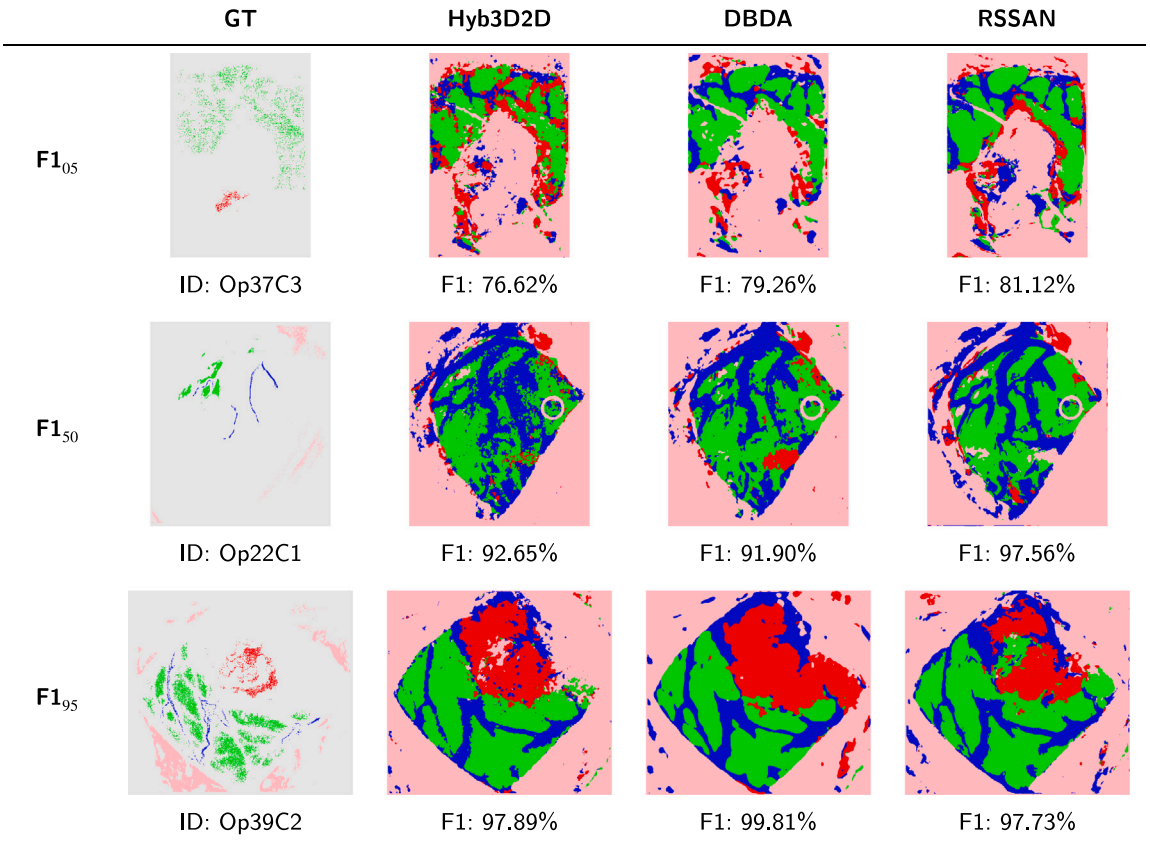


Fig. 9. Representative HELICoiD examples for the three best-performing models presented in Fig. 6. Rows $F1_{05}$, $F1_{50}$, and $F1_{95}$ denote examples selected from the 5th, 50th, and 95th percentiles of the per-image F1-score distribution. Colors indicate healthy tissue in green, tumor in red, blood vessels in blue, and background in pink.

related to a more complex and compact mapping performed by the spectral-spatial attention modules, whose separability is not fully reflected by the intermediate SDI measurements. However, this behavior is less evident for the lower-spectral-resolution SLIMBRAIN dataset, suggesting a poorer adaptation of RSSAN to low-spectral inputs.

The relevance of local feature extraction becomes clearer when considering the convolutional models. Hyb3D2D showed strong performance on both datasets, replicating the sharp increase in SDI metrics in the output logits seen with RSSAN. Its behavior can be better interpreted by considering the results obtained by EfficientNet, which combines DW convolutions along with squeeze-and-excitation (SE) blocks. In contrast, GhostNet, which also incorporates SE blocks but relies also on ghost modules and a different arrangement of pointwise and DW convolutions, performed notably worse. This suggests that DW convolutions may have played a more relevant role than SE blocks alone. However, the lower performance of LiteDwNet indicates that the use of DW convolutions is not sufficient by itself. In this case, the rapid reduction of spectral features in the input layer may have caused the loss of relevant information before subsequent processing stages. As Fig. 10 shows, the final separability observed in the output logits is less pronounced than that obtained by Hyb3D2D, possibly indicating a less effective utilization of the spectral and spatial information in both datasets. Conversely, the relatively high final SDI value observed for LiteDwNet can be associated with the AUC obtained on the SLIMBRAIN dataset, which is the second highest after Hyb3D2D. However, the simplified architecture and a potentially ineffective use of the multi-branch design may lead to a suboptimal transformation of the learned representations into probability estimates.

These observations also suggest that 3D convolutions are not necessarily superior for high-dimensional spectral data. Several well-performing models, including RSSAN and EfficientNet, rely mainly on 2D convolutions after the initial processing stages. At the same time, DBDA shows that an initial reduction of spectral information does not necessarily compromise performance when followed by specialized spatial and spectral branches. The comparison between DBDA and LiteDwNet is particularly informative in this respect, since both models use a similar spectral-oriented input kernel, but obtain substantially different results. This difference is likely related to the subsequent feature processing strategy adopted by DBDA, where two dedicated branches separately emphasize spectral and spatial information through dense convolutional blocks and attention modules.

A deeper analysis of the global F1-score points out that the key factor is not simply whether an architecture is single-branch or multi-branch, but whether spatial and spectral information are processed in a differentiated way. This can be observed by comparing HiT and SSAN. HiT integrates convolutional operations inside attention layers through conv-permutator blocks, where features are projected along height, width, and spectral dimensions before additive attention. SSAN, instead, explicitly separates a spectral-oriented stage based on $1 \times 1 \times 7$ convolutional kernels from a spatial-oriented stage combining attention layers and 3×3 convolutions. This more explicit specialization may explain the slightly better metrics obtained by SSAN. Following this reasoning, Fig. 6 confirms that all models that use attention layers but with differentiated spatial and spectral stages perform better than those models that do not. The only exception is RSSAN with SLIMBRAIN, for which its attention-based input header might not be suitable for low spectral resolution data.

4.2. Computational performance and accuracy-efficiency trade-off

Regarding computational performance, Fig. 7b and d show that HSI-specialized models consistently obtain lower inference times than

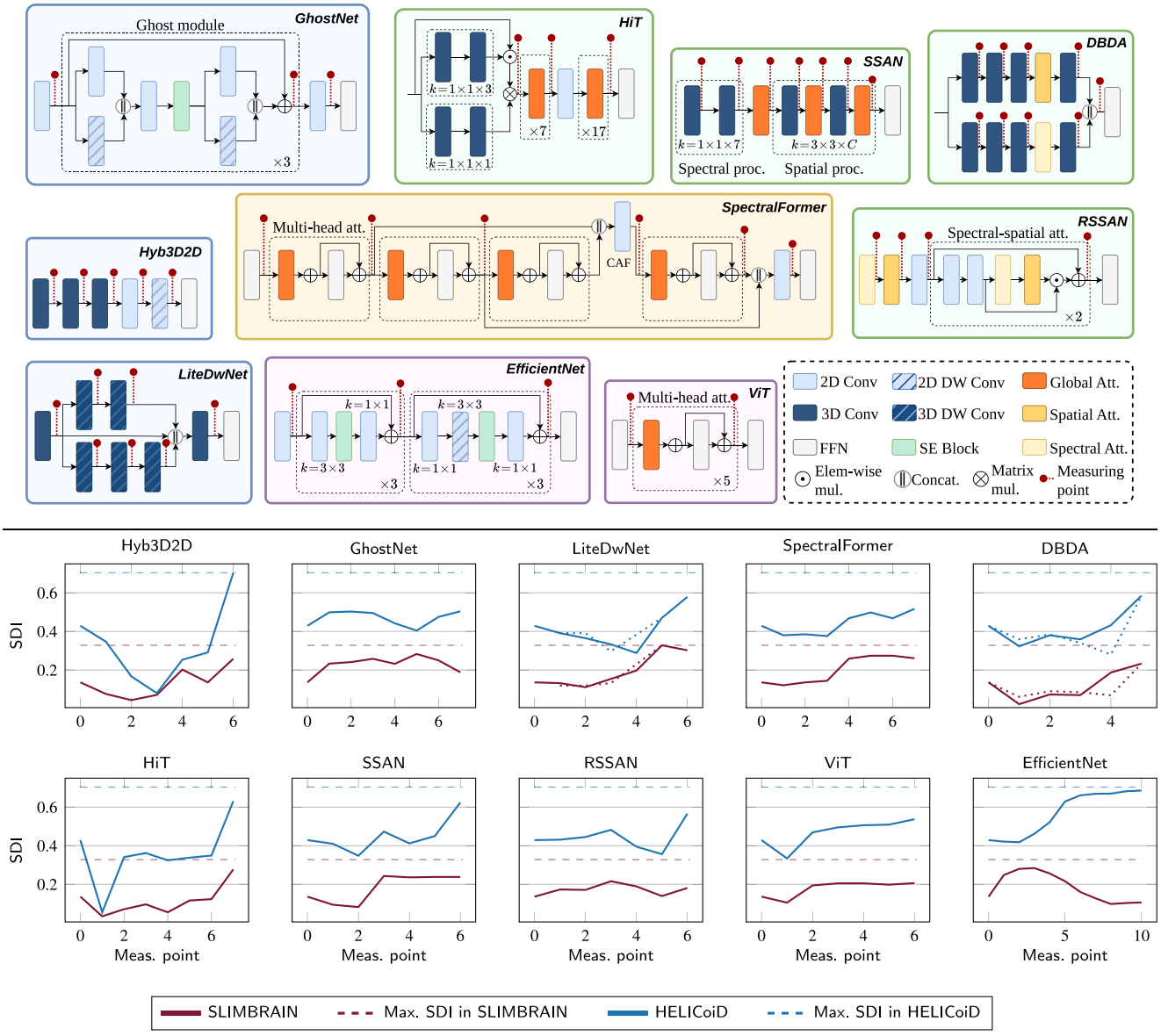


Fig. 10. Block diagrams of the studied architectures indicating, with red dots, the stages at which the SDI metric is computed (upper part). The lower part of the figure reports the SDI values obtained at each point for the SLIMBRAIN and HELICoiD datasets. The dotted lines in the SDI plots corresponding to the DBDA model indicate the values associated with the spatial branch, whereas in the LiteDwNet model they represent the left branch.

baseline models. This aspect is particularly relevant for clinical translation, since intraoperative HSI systems should provide timely predictions without requiring excessive computational infrastructure or disrupting the surgical workflow [56]. Therefore, architectures that reduce inference time while preserving competitive segmentation performance are more suitable candidates for clinically oriented deployment.

In this study, the observed efficiency is partly related to the way HSI-specific architectures reduce or reorganize spectral information in their input stages. Models such as LiteDwNet, DBDA, and HiT use large spectral kernels or spectral-oriented projections to reduce the computational burden of subsequent layers. RSSAN follows a different strategy, as its attention-based input header aggregates spatial and spectral information through dedicated attention modules.

At the same time, clinical translation cannot rely on computational efficiency alone, since fast inference is useful only when associated with reliable segmentation performance. Therefore, Fig. 7a and c also show a trade-off between computational load and classification performance in both datasets. Most models require more parameters to process the denser spectral information of HELICoiD. EfficientNet represents an exception, since its model size and inference time remain similar across datasets. This behavior is related to its 2D convolutional structure, where internal feature projections retain the same dimensionality after the input layer, independently of the original number of spectral bands.

Together, these results show that high segmentation performance does not necessarily correspond to favorable computational scalability. Therefore, for clinically oriented applications, model selection should follow a multidimensional evaluation strategy, such as the one adopted in this benchmark, jointly considering accuracy, inference time, and model complexity rather than optimizing classification performance alone.

4.3. Prediction variability and clinical implications

Beyond architectural and computational considerations, the results reveal substantial prediction variability across images, which is particularly relevant when considering the possible clinical use of HSI-based

segmentation models. This variability is especially evident for the tumor class in SLIMBRAIN, where all models except ViT exhibit standard deviations exceeding their corresponding mean tumor F1-score values. Such behavior indicates that average metrics alone may provide an incomplete description of model reliability: in some images, models detect tumor samples, whereas in others detection largely fails.

This instability is consistent with the exploratory dataset analysis. The t-SNE representations in Fig. 3 show limited separability of tumor samples, particularly in SLIMBRAIN, where tumor pixels largely overlap with healthy tissue and dura mater. Furthermore, the SNR analysis in Fig. 4 indicates that SLIMBRAIN has substantially lower SNR values than HELICoiD, in addition to a lower number of spectral bands. These factors likely reduce the discriminative information available to the models and contribute to the unstable tumor predictions observed in the results.

From a clinical perspective, this finding is important because unstable tumor segmentation can limit the trustworthiness of model-assisted decision support. In an intraoperative scenario, a model with acceptable average performance but highly variable tumor detection may be problematic, since failure cases could lead to missed tumor regions or over-segmentation into non-tumoral tissue. The former could increase the risk of leaving residual tumor tissue, whereas the latter could unnecessarily extend the resection area, potentially affecting healthy or functionally relevant tissue.

4.4. Limitations

The reported results should be interpreted in light of several methodological and dataset-related limitations. Although the PRISMA methodology allowed models to be selected for evaluation according to rigorous and reproducible criteria, this does not necessarily imply that the selected architectures are the best currently available. Some criteria, such as the citation count, favored more mature and established models, potentially excluding newer but promising approaches (e.g., state-space model architectures). Therefore, this benchmark should be interpreted as a systematic comparison of representative and well-documented architectures rather than as an exhaustive evaluation of all possible models usable for HSI-based brain tumor segmentation.

A second limitation concerns the domain shift between the original application context of several selected architectures and the medical HSI setting considered in this work. Most of the evaluated models were originally developed for remote sensing HSI datasets, which differ from intraoperative brain HSI in terms of spectral range, acquisition conditions, spatial structures, semantic classes, and sources of variability. In particular, remote sensing HSI typically spans broader spectral ranges, often including VNIR-SWIR (visible and near-infrared to shortwave infrared) information, and is generally associated with terrestrial materials such as vegetation, soil, minerals, or urban structures [57]. Conversely, brain HSI relies on more targeted spectral configurations, such as VNIR or NIR (near-infrared) bands, which are mainly related to physiological and pathological tissue properties, including oxygenation, blood volume, vascular patterns, and tumor margins [21]. Although this transfer was intentionally investigated in the present benchmark, some architectures may have been disadvantaged by the specific characteristics of medical data and might require domain-specific adaptation to fully exploit their potential. A related consideration applies to the RGB baseline models. Although they were included as reference architectures because of their widespread use in image analysis, they were originally designed for three broad visible channels rather than for high-dimensional spectral data. Their inclusion therefore provides a useful comparison point, but their behavior should be interpreted considering this difference in input representation.

A related limitation concerns the absence of model-specific hyperparameter optimization. The same training strategy, including learning rate, batch size, number of epochs, optimizer, and patch size, was used across architectures to preserve comparability. This may not reflect the best achievable performance of each model. Some architectures could benefit from tailored hyperparameter tuning, regularization strategies, or architecture-dependent training schedules.

The use of a fixed patch size for all architectures also represents a methodological constraint. This choice was made to ensure common test conditions and fair comparison across models. However, different architectures were originally designed and optimized using different spatial contexts, and a single patch size may not be optimal for all of them.

Another limitation is related to the sparse nature of the available GT annotations. Since only a subset of pixels was labeled, the evaluation was restricted to annotated regions and may not fully capture the behavior of models over the entire surgical scene. Moreover, although the morphological GT augmentation was applied to locally increase the number of labeled pixels, it may introduce annotation uncertainty.

Finally, although the number of images used in the two datasets was substantial compared to other past studies, their size remains limited with respect to typical DL requirements. Moreover, the datasets include subjects with different types of tumors, origins and grades, as well as images acquired under different conditions. This heterogeneity likely contributed to the variability observed across the experimental repetitions, particularly for tumor segmentation. Similar benchmark-to-clinical domain shifts have been reported in brain tumor segmentation, where differences in image characteristics and preprocessing can affect model generalization [58].

5. Conclusions

This work introduced a benchmark covering a broad number of recent DL architectures selected by following the PRISMA guidelines. A taxonomy was defined to categorize the selected architectures according to their main design mechanisms and spectral-spatial processing strategies. This taxonomy also facilitated a detailed analysis of the quantitative results, revealing a relationship between performance and the identified architectural arrangements. The evaluation was conducted on two different *in-vivo* brain tumor HS datasets: SLIMBRAIN, featuring 25 spectral bands, and HELICoiD with 128 spectral channels. The presented analysis highlighted the notable performance of models relying on depth-wise convolutions. Based on the metrics obtained, they may represent one of the most promising building blocks for designing NN architectures for HSI classification. Moreover, the model ranking obtained indicated that 3D convolutions are not necessarily superior to 2D convolutions.

Another important observation was that attention mechanisms achieved better results when applied separately to spatial and spectral features. However, the type of operation performed within each layer appears to be more critical than whether it is applied in parallel or sequentially.

Furthermore, although the best-performing models for both datasets are convolution-based, this did not clearly establish the superiority of convolutional architectures over hybrid configurations, as the performance similarity among the top five models in both datasets prevents a definitive conclusion. The analysis of the spectral characteristics of both datasets corroborated the experimental results: HELICoiD generally resulted in more favorable learning conditions for the tested architectures. However, analysis of inference time and model size revealed a trade-off between computational cost and the classification metrics obtained.

Ultimately, the findings of this study clarify how different architectural mechanisms behave in medical HSI, highlighting their performance and efficiency trade-offs and supporting a more informed understanding of future design choices.

CRediT authorship contribution statement

Guillermo Vazquez: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Domenico Ragusa:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Emanuele Torti:** Writing – review & editing, Validation, Supervision. **Elisa Marenzi:** Writing – review & editing, Validation, Supervision. **Eduardo Juarez:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Angel M. Groba:** Supervision, Resources, Project administration. **Francesco Leporati:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Ethical statement

Data from the SLIMBRAIN database were acquired in accordance with the Declaration of Helsinki guidelines and the study protocol was approved by the Research Ethics Committee of the Hospital Universitario 12 de Octubre, Madrid, Spain (protocol code 19/158, 28 May 2019). For the HELICoiD database, the study protocol and consent procedures were approved by the Research Ethics Committee of Doctor Negrín University Hospital (Ref 130069 for the first and second campaigns, and Ref 2019-001-1 for the third campaign), obtaining written informed consent from all participants, including the publication of potentially identifiable images or data. This study did not require additional ethical approval, as it used previously collected and fully anonymized datasets with prior ethical clearance and informed consent.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Guillermo Vazquez reports financial support was provided by European Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The *in-vivo* HS human brain data used in this study are available from the SLIMBRAIN database and the HELICoiD dataset at https://slimbrain.citsem.upm.es/ and https://hsibraindatabase.iuma.ulpgc.es/, respectively. Access must be granted by the data owners upon reasonable request. The source code used in this study is available on GitHub at https://github.com/UPM-UniPV/HSIBrain-Benchmark.

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